

Title

Safety, tolerability, and lipid response after short-term administration of TLC-2716 in humans

Abstract

Background: Selective targeting of hepatic lipid synthesis pathways offers a novel strategy to address residual cardiometabolic risk. **Methods:** We conducted a randomized, placebo-controlled phase 1 evaluation of TLC-2716 in adults with or at risk for dyslipidemia. Participants received 14 days of once-daily dosing. Outcomes included adverse event rates, laboratory abnormalities, and plasma lipoprotein concentrations. **Results:** A total of 28 subjects completed the trial. No serious adverse events were observed, and discontinuations due to tolerability were infrequent (8%). Triglyceride levels fell by an average of 31% in the treatment arm relative to baseline, while placebo showed minimal change. Postprandial remnant cholesterol declined by approximately 52% in the active group. Hepatic enzyme activities and glucose homeostasis remained stable. **Conclusion:** Short-term TLC-2716 exposure was well-tolerated with favorable lipid-modifying effects, supporting progression to Phase 2 efficacy studies.